Kushal Kafle

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Research Interests

Multimodal large language models, vision-language representation learning, large-scale foundation model training with web-scale data, benchmarks, evaluation and model probing, robustness and fairness

Selected Technology Transfers

I lead Adobe's vision-language foundation model efforts, spanning image-text embedding models and, since late 2024, multimodal large language models. Since 2023, the embedding models have powered features across many Adobe products:

- **Semantic Search in Express and Stock** *2023 – present*
  Natural-language and image search for templates in Adobe Express (+46% click-through) and for assets in Adobe Stock. It has since grown to multilingual, behavior-tuned, and steerable search, and now runs on a new generation of our foundation model.
- **Lightroom Natural-Language Search** *2025 – present*
  Search a photo library with full phrases like "photos of dog taken at the beach", with no keywords needed. Launched in Lightroom on the web, then in Lightroom desktop and mobile.
- **Adobe Firefly Image Generation** *2023 – 2025*
  Our text and image encoders powered the Firefly Image models' reference features, including Style Reference uploads, generating similar images, and using images as references.
- **Premiere Pro Media Intelligence** *2025*
  Visual search in Premiere Pro's Media Intelligence and Search panel, built with the video team on our image-text foundation model.
- **Other transfers:** Firefly image-to-video (prompt coherence and safety); Firefly NSFW safety filter; AEM Smart Tags; Lightroom assisted culling and Auto Stack (on-device); Express "More like this" template similarity; Photoshop prompt-free object compositing (beta); Firefly training-data filtering models.

Research Experience

Mar 2020-Current	**Adobe Research**, San Jose, CA **Senior Research Scientist 2** (Feb 2026 – present) Senior Research Scientist (Jan 2024 – Feb 2026) Research Scientist 2 (Feb 2022 – Jan 2024) Research Scientist (Mar 2020 – Feb 2022) **Manager**: Scott Cohen **Responsibilities**: I lead Adobe's vision-language foundation models across the full lifecycle (data, legal, training infrastructure, novel algorithms, codebase development, and tech transfer): image-text embedding models that power features in Firefly, Express, Stock, Lightroom, and Premiere Pro, and, since late 2024, multimodal large language models for visual understanding. I also mentor research interns.
2019	**Research Intern**, Microsoft Research (w/ D. Florencio): scene-text entity recognition.
2017–18	**Research Intern**, Adobe Research (w/ S. Cohen, B. Price): DVQA and PReFIL.

Education

Aug 2014 - Feb 2020	**Ph.D. in Imaging Science** Chester F. Carlson Center for Imaging Science Rochester Institute of Technology, Rochester, NY **Advisor**: Dr. Christopher Kanan | **Research Group**: klab **Thesis Title**: Advancing Multi-modal Deep Learning: Towards Language-grounded Visual Understanding

Updated Nov 2026

Professional Services

Area Chair: WACV 2024, WACV 2025, ICCV 2025 (Lead AC), WACV 2026, CVPR 2026 (Lead AC)

Senior Program Committee: AAAI 2027

Journal Editing: Special Topic on *Identifying, Analyzing, and Overcoming Challenges in Vision and Language Research*, cross-posted to: Frontiers in Big Data, Frontiers in Robotics and AI, and Frontiers in Artificial Intelligence. Associate Editor.

Journal Reviewing: TPAMI (2019), TIP (2019), CVIU (2017-18), ACM-MM (2018-2019), TMI (2020), TDKE (2019), PR (2021), TMLR (2022-2024)

Conference Reviewing: CVPR (2017-2025), NeurIPS (2016-2025), AAAI (2017-2020), ICCV (2019,2023,2024), ICML (2020-2023), EMNLP (2019-2021), ACL (2020), NAACL (2021), ICLR (2022,2024), ECCV (2026). **Outstanding Reviewer Award** in CVPR 2018, 2019, 2020 and 2021, NeurIPS 2021, and ECCV 2026

Workshop Organization

- Workshop on shortcomings in vision and language (SiVL) at **ECCV, 2018**
- Workshop on shortcomings in vision and language (SiVL) **At NAACL, 2019**
- Drawings and abstract Imagery: Representation and Analysis (DIRA) **At ECCV, 2022**

Publications

Peer Reviewed International Conferences

(C1) Fioresi, J., Caba Heilbron, F., Nathani, P., Shah, M., & **Kafle, K.** (2026). Controlling Embedding Spaces with Text-Conditioned Transformations. *European Conference on Computer Vision (ECCV 2026)*.

(C2) Sethuraman, T. V., Khosla, S., Susladkar, O., Tiwari, A., Oh, S. W., **Kafle, K.**, Lee, J.-Y., Hoiem, D., & Jenni, S. (2026). ViTAL-X: Video-Text Alignment with Cross-Modal Temporal Edits. *European Conference on Computer Vision (ECCV 2026)*.

(C3) Akdemir, K., Shi, J., **Kafle, K.**, Price, B., & Yanardag, P. (2026). Plot'n Polish: Zero-Shot Story Visualization and Disentangled Editing with Text-to-Image Diffusion Models. *AAAI Conference on Artificial Intelligence (AAAI 2026)*.

(C4) Hernandez, J., Shi, J., Jenni, S., Ordonez, V., & **Kafle, K.** (2025). Improving Large Vision and Language Models by Learning from a Panel of Peers. *IEEE/CVF International Conference on Computer Vision (ICCV 2025)*.

(C5) Slyman, E., Tanjim, M., **Kafle, K.**, & Lee, S. (2025). Calibrating MLLM-as-a-judge via Multimodal Bayesian Prompt Ensembles. *IEEE/CVF International Conference on Computer Vision (ICCV 2025)*.

(C6) Khosla, S., Tiwari, A., **Kafle, K.**, Jenni, S., Zhao, H., Collomosse, J., & Shi, J. (2025). MAGNET: Augmenting Generative Decoders with Representation Learning and Infilling Capabilities. *Annual Meeting of the Association for Computational Linguistics (ACL 2025)*.

(C7) Hua, H., Shi, J., **Kafle, K.**, Jenni, S., Zhang, D., Collomosse, J., Cohen, S., & Luo, J. (2024). FineMatch: Aspect-Based Fine-Grained Image and Text Mismatch Detection and Correction. *European Conference on Computer Vision (ECCV 2024)*.

(C8) Slyman, E., Lee, S., Cohen, S., & **Kafle, K.** (2024). FairDeDup: Detecting and Mitigating Vision-Language Fairness Disparities in Semantic Dataset Deduplication. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)*.

(C9) Sameni, S., **Kafle, K.**, Tan, H., & Jenni, S. (2024). Building Vision-Language Models on Solid Foundations with Masked Distillation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)*.

(C10) Yang, Z., **Kafle, K.**, Lin, Z., Cohen, S., Ding, Z., & Ordonez, V. (2024). SCoRD: Subject-Conditional Relation Detection with Text-Augmented Data. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2024)*.

(C11) Tanjim, M. M., Singh, K. K., **Kafle, K.**, Sinha, R., & Cottrell, G. W. (2024). Discovering and Mitigating Biases in CLIP-based Image Editing. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2024)*. Oral Presentation (2.6% of submissions)

(C12) Yang, Z., **Kafle, K.**, Dernoncourt, F., & Ordonez, V. (2023). Improving Visual Grounding by Encouraging Consistent Gradient-Based Explanations. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023)*.

(C13) Tanjim, M. M., Singh, K. K., **Kafle, K.**, Sinha, R., & Cottrell, G. W. (2022). Debiasing Image-to-Image Translation Models. *British Machine Vision Conference (BMVC 2022)*.

(C14) Pham, K., **Kafle, K.**, Lin, Z., Ding, Z., Cohen, S., Tran, Q., & Shrivastava, A. (2022). Improving Closed and Open-Vocabulary Attribute Prediction Using Transformers. *European Conference on Computer Vision (ECCV 2022)*.

(C15) Shrestha, R., **Kafle, K.**, & Kanan, C. (2022). OccamNets: Mitigating Dataset Bias by Favoring Simpler Hypotheses. *European Conference on Computer Vision (ECCV 2022)*. Oral Presentation (2.7% of submissions)

(C16) Shrestha, R., **Kafle, K.**, & Kanan, C. (2022). An Investigation of Critical Issues in Bias Mitigation Techniques. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2022)*.

(C17) Ravi, H., **Kafle, K.**, Cohen, S., Brandt, J., & Kapadia, M. (2021). AESOP: Abstract Encoding of Stories, Objects, and Pictures. *IEEE/CVF International Conference on Computer Vision (ICCV 2021)*.

(C18) Pham, K., **Kafle, K.**, Lin, Z., Ding, Z., Cohen, S., Tran, Q., & Shrivastava, A. (2021). Learning to Predict Visual Attributes in the Wild. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2021)*.

(C19) Teney, D., **Kafle, K.**, Shrestha, R., Abbasnejad, E., Kanan, C., & van den Hengel, A. (2020). On the Value of Out-of-Distribution Testing: An Example of Goodhart's Law. *Advances in Neural Information Processing Systems (NeurIPS 2020)*.

(C20) Shrestha, R., **Kafle, K.**, & Kanan, C. (2020). A Negative Case Analysis of Visual Grounding Methods for VQA. *Annual Meeting of the Association for Computational Linguistics (ACL 2020)*.

(C21) Hayes, T. L.*, **Kafle, K.***, Shrestha, R.*, Acharya, M., & Kanan, C. (2020). REMIND Your Neural Network to Prevent Catastrophic Forgetting. *European Conference on Computer Vision (ECCV 2020)*. *equal contribution

(C22) **Kafle, K.**, Shrestha, R., Price, B., Cohen, S., & Kanan, C. (2020). Answering Questions about Data Visualizations using Efficient Bimodal Fusion. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2020)*.

(C23) Shrestha, R., **Kafle, K.**, & Kanan, C. (2019). Answer Them All! Toward Universal Visual Question Answering Models. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2019)*.

(C24) Acharya, M., **Kafle, K.**, & Kanan, C. (2019). TallyQA: Answering Complex Counting Questions. *AAAI Conference on Artificial Intelligence (AAAI 2019)*.

(C25) **Kafle, K.**, Price, B., Cohen, S., & Kanan, C. (2018). DVQA: Understanding Data Visualizations via Question Answering. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2018)*.

(C26) **Kafle, K.** & Kanan, C. (2017). An Analysis of Visual Question Answering Algorithms. *IEEE/CVF International Conference on Computer Vision (ICCV 2017)*.

(C27) **Kafle, K.**, Yousefhussien, M., & Kanan, C. (2017). Data Augmentation for Visual Question Answering. *International Conference on Natural Language Generation (INLG 2017)*.

(C28) **Kafle, K.** & Kanan, C. (2016). Answer-Type Prediction for Visual Question Answering. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2016)*.

Peer Reviewed Journals

(J1) **Kafle, K.**, Shrestha, R., & Kanan, C. (2019). Challenges and Prospects in Vision and Language Research. *Frontiers in Artificial Intelligence*.

(J2) **Kafle, K.** & Kanan, C. (2017). Visual Question Answering: Datasets, Algorithms, and Future Challenges. *Computer Vision and Image Understanding (CVIU)*.

Workshop Papers

(W1) Lu, J., Srivastava, S., Chen, J., Shrestha, R., Acharya, M., **Kafle, K.**, & Kanan, C. (2025). Revisiting Multi-Modal LLM Evaluation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2025)*.

Preprints

(P1) Abdel Magid, S., Chun, S., Slyman, E., **Kafle, K.**, & Russakovsky, O. (2026). Copying As Defense: Protecting Images from Editing By Eliciting Model Copying. *arXiv preprint* arXiv:2610.35566.

(P2) Lu, J., Jenni, S., **Kafle, K.**, Shi, J., Zhao, H., & Fu, Y. (2026). Seeing Through Words: Controlling Visual Retrieval Quality with Language Models. *arXiv preprint* arXiv:2602.21175.

(P3) Wu, Q., Shi, J., Jenni, S., **Kafle, K.**, Wang, T., Chang, S., & Zhao, H. (2026). RetouchIQ: MLLM Agents for Instruction-Based Image Retouching with Generalist Reward. *arXiv preprint* arXiv:2602.17558.

(P4) Just, H. A., Fan, Y., Zhao, H., Gu, J., Zhang, R., Jenni, S., **Kafle, K.**, Jia, R., & Shi, J. (2025). More Than the Final Answer: Improving Visual Extraction and Logical Consistency in Vision-Language Models. *arXiv preprint* arXiv:2512.12487.

(P5) Dong, Q., Figueroa, L., Zhao, H., **Kafle, K.**, Kuen, J., Ding, Z., Cohen, S., & Fu, Y. (2025). CoT Referring: Improving Referring Expression Tasks with Grounded Reasoning. *arXiv preprint* arXiv:2510.06243.

(P6) Zang, Y., Tan, H., Yoon, S., Dernoncourt, F., Gu, J., **Kafle, K.**, Sun, C., & Bui, T. (2025). MS4UI: A Dataset for Multi-modal Summarization of User Interface Instructional Videos. *arXiv preprint* arXiv:2506.12623.

(P7) Abdel Magid, S., Wang, J.-H., **Kafle, K.**, & Pfister, H. (2024). They're All Doctors: Synthesizing Diverse Counterfactuals to Mitigate Associative Bias. *arXiv preprint* arXiv:2406.11331.

(P8) Qian, Z., Hayes, T. L., **Kafle, K.**, & Kanan, C. (2020). Do We Need Fully Connected Output Layers in Convolutional Networks? *arXiv preprint* arXiv:2004.13587.

Invited Talks and Guest Lectures

- **Understanding and mitigating model bias in the age of large models** Keynote, BISCUIT Workshop at ICCV 2025 (Oct 2025)
- **Why bias affects machine learning and what can we do about it?** The 2021 SIGIR Workshop On eCommerce, ECOM@SIGIR (July 2021)
- **Vision Language Models and VQA:** 3rd Nepal Winter School in AI (Dec 2021)
- **Biases in Vision and Language: Visual Question Answering**
  - Guest Lecture for the Virginia Tech class *CS 6501/4501: Vision and Language* (Sept 2020)
  - Guest Lecture for the RIT Class *CS 6501/4501: Vision and Language* (Sept 2020)
- **Language Grounded Visual Understanding**: Microsoft, and Adobe Research (Nov 2019)

Skills

- **Technical**: PyTorch, Python, large-scale distributed training, billion-scale data pipelines and filtering, Git, LaTeX, Bash, AWS
- **Data**: Scraping and curating web-scale data, data licensing and legal review, vendor and crowd-sourced annotation (MTurk, Appen, LabelBox, Hive, iMerit)